

1. Scenario: Event Communication Security (Caesar Cipher – Shift of 4)

a. Encrypt the message "MEET AT NOON" using a shift of 4

Rule: Shift each letter **4 positions forward** in the alphabet.

Plaintext M E E T A T N O O N

Ciphertext Q I I X E X R S S R

Encrypted Message:

QIIX EX RSSR

b. How does the authorized member retrieve the original message?

The receiver knows that the message was encrypted using a **Caesar Cipher with a shift of 4**. They simply shift each letter **4 positions backward**.

Example:

- $Q \rightarrow M$
- $I \rightarrow E$
- $X \rightarrow T$

After shifting all letters back by 4 positions, the original message is recovered:

MEET AT NOON

2. Scenario: Intercepted Suspicious Message

Ciphertext:

GSRH RH Z HVXIVG

This is encrypted using the **Atbash Cipher**, where:

- $A \leftrightarrow Z$
- $B \leftrightarrow Y$
- $C \leftrightarrow X$
- ...

Decrypting:

Cipher	G	S	R	H		R	H		Z		H	V	X	I	V	G	
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Plain	T	H	I	S		I	S		A		S	E	C	R	E	T	

Original Message

THIS IS A SECRET

Justification

- The repeating word **RH** appears twice, suggesting a common word like **IS**.
- The Atbash cipher substitutes each letter with its opposite in the alphabet.
- Applying the reverse substitution gives the meaningful sentence "**THIS IS A SECRET.**"

3. Scenario: Secure Messaging System (Vigenère Cipher)

Plaintext

HELLO

Keyword

KEY

Repeat the keyword until it matches the plaintext length:

Plaintext H E L L O

Keyword K E Y K E

Convert letters to numbers:

Letter Value

H	7
E	4
L	11
L	11
O	14
K	10
E	4

Letter Value

Y 24

Encryption Formula:

$$\text{Cipher} = (\text{Plain} + \text{Key}) \bmod 26$$

Plain	Key	Calculation	Cipher
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H(7)	K(10)	7+10=17	R
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E(4)	E(4)	4+4=8	I
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L(11)	Y(24)	35 mod 26 = 9	J
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L(11)	K(10)	21	V
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O(14)	E(4)	18	S
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Ciphertext

RIJVS

Role of the Repeating Keyword

- The keyword **KEY** repeats as **KEYKE**.
- Each keyword letter determines how much the corresponding plaintext letter is shifted.
- Because the shift changes for every character, the Vigenère cipher is more secure than the Caesar cipher.

4. Scenario: Suspicious Digital Asset Investigation (Steganography)

How investigators detect hidden data

Investigators can:

1. Perform Statistical Analysis

- Analyze pixel values for unusual patterns.
- Detect changes in Least Significant Bits (LSB).

2. Use Steganalysis Tools

- Tools such as **StegExpose**, **StegSolve**, or **zsteg** can identify hidden information.
- Compare the suspicious image with the original version.

Two techniques to improve security

1. Encrypt the secret message before embedding

- Even if extracted, the hidden data remains unreadable without the encryption key.

2. Randomized or Adaptive Embedding

- Hide data in randomly selected pixels using a secret key instead of sequential pixels.
 - This makes detection much more difficult.
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5. Scenario: Legacy Encryption in Corporate Communication (Rail Fence Cipher)

a. Original Message

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

This is a **Rail Fence Cipher (3 rails)**.

Decrypted Message

WE ARE DISCOVERED FLEE AT ONCE

b. Why is this method vulnerable?

- It only rearranges letters without changing them.
 - It has **no encryption key** (or a very small key space).
 - Modern computers can easily try all possible rail numbers.
 - Frequency analysis and brute-force attacks can quickly reveal the original message.
 - It provides weak security for modern communications.
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c. One practical enhancement

Encrypt the message using a modern encryption algorithm such as AES before transmission.

OR

Combine transposition with a strong substitution cipher (double encryption).

This significantly improves confidentiality because attackers must break both encryption layers.